

Path-Aware Positional Encoding for Raster-Scan Autoregressive Image Generation in CRT Signal Space

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Abstract—We study autoregressive image generation in a 1D *signal space* obtained by following a CRT-inspired raster-scan path with a differentiable Gaussian-beam renderer. Within this framing, the sequence position of a sample and its 2D spatial position on the screen are not independent: the path is a deterministic, one-to-one function of the sequence index. We show empirically that, given a sinusoidal 2D path positional encoding, adding the standard 1D sequence positional encoding consistently *harms* test likelihood. On Oxford Flowers 102 at 32×32 , removing the sequence encoding improves test bits-per-pixel from 7.16 ± 0.06 to 6.43 ± 0.18 across three seeds at 30 epochs (Welch $t = 5.4$). We also provide multi-seed results for all four positional-encoding configurations, finding that the sequence encoding alone (without path) yields 7.12 ± 0.11 bpd, nearly identical to the full model, while removing both encodings degrades performance to 9.03 ± 0.01 bpd, confirming that the path encoding is the informative component. The direction of the effect is preserved in a three-seed comparison at 48×48 (Welch $t = -1.97$, 15 epochs). Extending this analysis at a matched sequence length ($N=1024$, single-sample-per-pixel raster), we find that *traversal locality* – the mean sequence distance between four-neighbour pixel pairs – predicts test bpd with Pearson $r = 0.996$ ($p < 10^{-4}$, $n = 9$) across four hand-designed traversals and five random pixel permutations, identifying 1D serialisation of 2D structure as the architectural bottleneck rather than the specific scan order. Four architectural probes (delta input encoding, patch-hierarchical traversal, epoch-level traversal augmentation, and a causal 1D conv-stem) all fail to improve over the matched- N raster baseline, suggesting the remaining gap to pixel-space methods requires genuinely 2D-aware computation rather than surface-level re-orderings or shallow 1D inductive bias. We additionally test three CRT-physics-motivated architectural modifications (learned per-step beam focus, phosphor persistence, and learned beam-position perturbation) against the same baseline at three seeds; none reaches statistical significance ($|t| < 1$). A matched-parameter PixelCNN++ trained under the same protocol attains a stronger absolute likelihood (4.71 ± 0.02 bpd at 30 epochs), so the observed encoding effect is best read as architecture-specific to signal-space autoregressive models rather than a closure of the gap to pixel-space methods.

Index Terms—autoregressive models, image generation, positional encoding, likelihood, ablation

I. INTRODUCTION

Autoregressive image models factor a joint distribution $p(\mathbf{x})$ over an image $\mathbf{x}$ as a product of conditional distributions

over a chosen ordering of its components. PixelRNN and PixelCNN [1] introduced the now-standard choice of a row-major raster scan over pixels, and subsequent work [2], [3], [4] has retained that ordering while replacing the underlying architecture. Within this lineage, position information is uniformly encoded by the position in the 1D sequence, which is identified with the position in the 2D image through the implicit raster mapping.

A separate body of work [5], [6], [7], [8], [9] treats image synthesis as the production of a continuous trajectory that is then deposited onto a canvas by a (typically differentiable) renderer. The model never directly emits pixel intensities; it emits control signals for an external rendering process. We refer to this family as *signal-space* generative modelling.

The two framings can be unified by adopting an *intermediate* representation: the model emits a 1D sequence of intensities sampled along a deterministic raster-scan path, and a Gaussian-beam renderer deposits those intensities onto a 2D grid. This formulation makes explicit a property that is implicit in PixelRNN-style models: each 1D sequence position carries an associated 2D path coordinate, and the mapping between them is invertible. As a result, the modeller has two natural choices for positional encoding, the 1D sequence index and the 2D path coordinate, which are individually unsurprising but *redundant* when used together.

This paper studies the consequences of that redundancy. We find that it is not benign: at matched parameters and training budget, adding the 1D sequence encoding to a model that already has a 2D path encoding makes test likelihood reliably worse. We verify this with multi-seed ablations at two image sizes, examine three CRT-physics-motivated architectural modifications as additional ablation targets, and benchmark against a matched-parameter PixelCNN++ to place the absolute numbers in context.

a) Contributions.:

- 1) We formalise the redundancy between sequence and path positional encodings in raster-scan signal-space autoregressive models, and show that adding the sequence encoding to a model that already has the path encoding hurts likelihood by (0.73 ± 0.19) bpd on Oxford Flowers

at 32×32 ($t = 5.4$, 3 seeds, 30 epochs). We additionally provide all four positional-encoding configurations at three seeds, confirming that the path encoding is the informative component. The direction of the effect persists at 48×48 (Welch $t = -1.97$, 3 seeds, 15 epochs).

- 2) We test three CRT-physics-motivated modifications (learned beam focus, phosphor persistence, beam-position perturbation) under the same protocol as a falsifiability check on the broader “CRT-inspired” framing. None reaches statistical significance ($|t| < 1$, three seeds each).
- 3) We provide a matched-parameter PixelCNN++ baseline under identical conditioning, training, and evaluation, allowing a direct comparison between signal-space and pixel-space autoregressive models on the same data.
- 4) We show that traversal locality predicts test bpd with Pearson $r = 0.996$ ($p < 10^{-4}$) across four hand-designed traversals and five random pixel permutations at matched $N=1024$, localising the bottleneck in serialised image AR to 1D-serialisation of 2D-neighbour structure rather than scan-order choice.
- 5) We test four architectural probes (delta input encoding, patch-hierarchical traversal, traversal augmentation, causal 1D conv-stem) against the matched- N raster baseline; none improves over baseline, narrowing the design space for future work to 2D-aware computational blocks.

II. RELATED WORK

a) Autoregressive image models.: PixelRNN/PixelCNN [1] established raster-order factorisation as the dominant approach. PixelCNN++ [2] introduced the discretised mixture-of-logistics (DMoL) output distribution and several other refinements; we adopt the DMoL head verbatim. Image Transformer [3] and iGPT [4] replace masked convolutions with self-attention while keeping the raster ordering. Recent work has revisited the ordering choice itself: RadAR [12] uses a radial scan, and [13] explores diagonal orderings, both reporting that raster order remains a strong default. None of these works isolates the role of the 1D sequence positional encoding given an additional 2D path encoding.

b) Stroke- and trajectory-based generation.: SketchRNN [5] models pen trajectories with an RNN. SPIRAL [6] learns a painting agent through reinforcement learning against a differentiable renderer. Paint Transformer [7] predicts strokes feed-forward. CLIPDraw [8] and VectorFusion [9] optimise SVG strokes using CLIP-based objectives. Our renderer is closest in spirit to the DiffVG [10] family but specialised to a raster scan with finite beam width.

c) Differentiable rasterisation.: The 2D Gaussian splatting view has a long history; a recent high-profile incarnation is 3D Gaussian splatting [11]. Our 2D variant is a special case with deterministic, fixed splat positions and a learnable (in V6 ablation) or fixed (default) beam width.

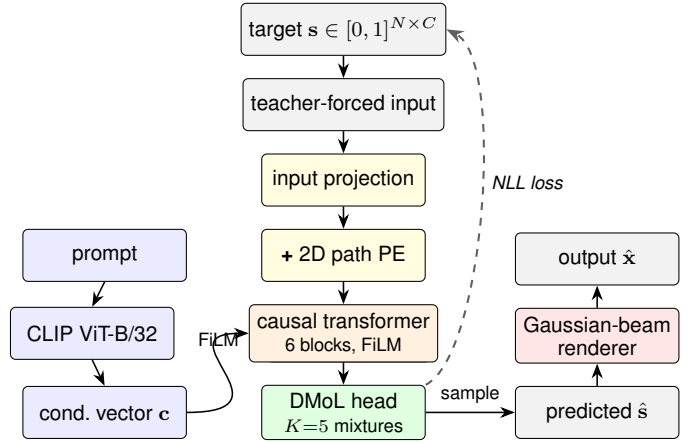


Fig. 1. Signal-space autoregressive architecture. The target image is converted to a 1D signal $\mathbf{s}$ by sampling along a fixed raster path (omitted from the figure for clarity). A causal transformer with FiLM conditioning on a CLIP text embedding $\mathbf{c}$ predicts the next signal value at each step. The DMoL output head provides the training likelihood (dashed arrow back to the target). At inference, the predicted signal $\hat{\mathbf{s}}$ is mapped to an image $\hat{\mathbf{x}}$ by the differentiable Gaussian-beam renderer (right column).

III. METHOD

We describe the signal-space framing (Section III-A), the architecture used to predict the signal (Section III-B), and the differentiable renderer that maps the signal back to a 2D image (Section III-C). We then define the four positional-encoding configurations that are the central object of study (Section III-D).

A. Signal-Space Framing

Let $\mathbf{x} \in [0, 1]^{C \times H \times W}$ be a target image. We fix a deterministic raster path $\mathcal{P} = (\mathbf{p}_1, \dots, \mathbf{p}_N)$ with $\mathbf{p}_i = (x_i, y_i) \in \mathbb{R}^2$ that traverses the image row by row at r samples per pixel, with a fraction f of every row allocated to a non-emitting flyback segment from the right edge back to the left edge. Each sample is assigned a binary mask $m_i \in \{0, 1\}$ with $m_i = 1$ on the active row sweep and $m_i = 0$ during flyback. For Oxford Flowers at 32×32 with $r = 2$ and $f = 0.08$ this yields $N = 2208$ and $\sum_i m_i / N \approx 0.928$.

The target signal $\mathbf{s} \in [0, 1]^{N \times C}$ is obtained by bilinearly sampling $\mathbf{x}$ at each path position:

$$\mathbf{s}_i = m_i \cdot \text{Bilinear}(\mathbf{x}, \mathbf{p}_i). \quad (1)$$

We model the joint distribution $p_\theta(\mathbf{s} | \mathbf{c})$, where $\mathbf{c}$ is a CLIP ViT-B/32 text embedding of a class-template caption such as “a photo of a [class name].”

B. Architecture

The architecture (Fig. 1) is a six-block causal transformer with FiLM [14] conditioning on the CLIP embedding and a DMoL output head with $K = 5$ mixture components. We use $d_{\text{model}} = 256$, $n_{\text{heads}} = 4$, feed-forward expansion factor 4, and dropout 0.1. The full model contains 6 520 867 trainable parameters.

For each step i the head emits parameters of an RGB-factorised mixture of logistics. We compute the negative log-likelihood $\text{NLL}(\mathbf{s})$ of the discretised target (256 bins per channel) and average over active positions:

$$\mathcal{L}(\theta) = \frac{\sum_i m_i \text{NLL}_\theta(\mathbf{s}_i)}{\sum_i m_i}. \quad (2)$$

Following [2] we train with teacher forcing, adding scheduled sampling [17] at probability p_{ss} linearly ramped from 0 to 0.25 across training to mitigate exposure bias on long sequences.

C. Differentiable Renderer

The renderer maps a predicted signal $\hat{\mathbf{s}}$ to a 2D image $\hat{\mathbf{x}} \in [0, 1]^{C \times H \times W}$ by depositing each sample as a 2D Gaussian at its path position with width σ :

$$\hat{\mathbf{x}}_{c,h,w} = \frac{1}{Z} \sum_{i=1}^N m_i \hat{\mathbf{s}}_{i,c} \exp\left(-\frac{(h-y_i)^2 + (w-x_i)^2}{2\sigma^2}\right), \quad (3)$$

where $Z = r \cdot 2\pi\sigma^2$ is a fixed normalisation tied to the kernel integral and the active-sample density. Unlike per-image max normalisation, this is smooth in the gradient and preserves a linear correspondence between signal magnitude and image brightness. We use $\sigma = 0.75$ throughout the main results.

D. Positional Encodings

The encoder optionally adds, at each step, two sinusoidal embeddings:

a) *1D sequence encoding* $\mathbf{e}_i^{\text{seq}}$: The standard [15] sinusoidal embedding indexed by $i \in \{1, \dots, N\}$.

b) *2D path encoding* $\mathbf{e}_i^{\text{path}}$: A 2D sinusoidal embedding of the continuous coordinate $\mathbf{p}_i = (x_i, y_i)$, normalised to $[0, 1]$. Half of the d_{model} channels encode x_i , the other half y_i .

We study four configurations: **both** (sequence + path), **seq only**, **path only**, and **neither**. Because the path is deterministic, $\mathbf{e}_i^{\text{path}}$ is recoverable from $\mathbf{e}_i^{\text{seq}}$ by composition with $\mathcal{P}$. The two encodings are therefore informationally redundant when used together: the question is whether their *form* hurts optimisation.

IV. EXPERIMENTS

A. Setup

a) *Dataset*: Oxford Flowers 102 [16] (7169 train / 1020 test images, all resized to 32×32 for the main experiments and 48×48 for the scaling study).

b) *Training*: AdamW [18] ($\beta_1 = 0.9$, $\beta_2 = 0.95$, weight decay 10^{-2}) with peak learning rate 3×10^{-4} , linear warmup over 500 steps and cosine decay to zero. Gradient clipping at norm 1.0. Mixed-precision training (fp16) on a single NVIDIA T4 GPU.

TABLE I
TEST BITS-PER-PIXEL AND PER-IMAGE NLL ON OXFORD FLOWERS
 32×32 , 30 EPOCHS, 3 SEEDS (SEEDS 0–2). LOWER IS BETTER.

Configuration	NLL (nats/img)	bpd
seq + path (full)	$15\,249.9 \pm 118.5$	7.156 ± 0.056
seq only	$15\,158.0 \pm 236.5$	7.119 ± 0.111
path only	$13\,695.9 \pm 389.8$	6.432 ± 0.183
neither	$19\,227.0 \pm 15.9$	9.030 ± 0.007

c) *Evaluation*: We report *bits-per-pixel* (bpd) under the same convention as PixelCNN++ [2]:

$$\text{bpd} = \frac{1}{HWC \ln 2} \mathbb{E}_{\mathbf{x}}[-\log p_\theta(\mathbf{x} | \mathbf{c})]. \quad (4)$$

Note that for the signal-space model the per-image NLL is the sum over all active signal positions, but the divisor in Eq. (4) is HWC (the number of sub-pixels), *not* the number of signal samples; this keeps the unit comparable across architectures.

B. Main Result: Sequence vs. Path Encoding

We compare the four positional-encoding configurations of Section III-D on the held-out test split. Table I reports mean and standard deviation across three seeds at 30 epochs.

With all four configurations now measured at three seeds, the bpd hierarchy is: **path only** (6.43 ± 0.18) < **seq only** (7.12 ± 0.11) $\approx$ **full** (7.16 ± 0.06) $\ll$ **neither** (9.03 ± 0.01). Two results stand out. First, the path-only configuration improves over full by 1554 nats/image (0.73 bpd); the pooled standard error is $\sqrt{(118.5^2 + 389.8^2)/3} = 235.5$ nats, giving $t = 6.6$ (Welch $t = 5.4$), a robust finding. Second, the absence of any encoding (“neither”) degrades performance drastically by 2.60 bpd relative to path-only, while seq-only alone recovers most of path-only’s gain (7.12 vs 9.03), confirming that the *path* encoding is the informative component. The near-identical performance of “full” and “seq only” ($t = -0.60$, not significant) shows that the sequence encoding adds negligible information beyond what the path encoding already provides, but does introduce a small optimisation penalty when both are present.

a) *Convergence study*: A single-seed 60-epoch run with the path-only configuration shows that the model continues to improve well past the 30-epoch budget (Fig. 2), reaching 5.24 bpd at the best checkpoint (epoch 55).

C. Scaling to 48×48

To check that the encoding effect is not specific to a particular sequence length, we repeat the comparison at 48×48 , where $N = 4944$, using three seeds at 15 epochs (Table II). The path-only configuration still outperforms the full model on average by 0.40 bpd (Welch $t = -1.97$). The effect is weaker than at 32×32 in statistical terms, mainly because one path-only seed achieved 8.03 bpd while the other two converged to ≈ 8.64 bpd, inflating the variance. The sign of the effect is nonetheless preserved across all three seeds, and the mean path-only bpd (8.43) is lower than the mean full bpd (8.83) by a consistent margin. We report this as supporting evidence that

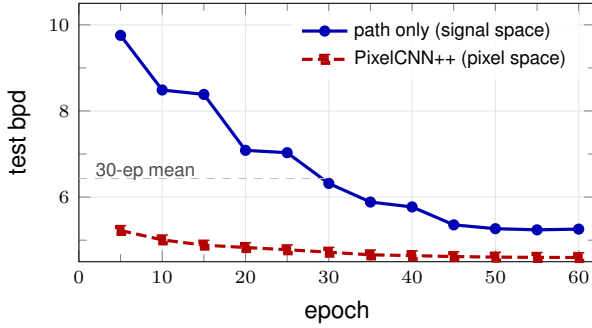


Fig. 2. Validation bits-per-pixel during training. Signal-space (blue) is the path-only configuration of Section IV-B; pixel-space (red, dashed) is the matched-parameter PixelCNN++ baseline of Section IV-D. Both are single-seed runs over 60 epochs. The dashed grey line marks the multi-seed mean of the path-only configuration at 30 epochs (6.43 bpd).

TABLE II

SCALING TO 48×48 : 15 EPOCHS, 3 SEEDS (SEEDS 0–2). THE PATH-ONLY ADVANTAGE PERSISTS WITH LONGER SEQUENCES (WELCH $t = -1.97$).

Configuration	bpd mean	bpd std
seq + path (full)	8.831	0.036
path only	8.433	0.348
Δ (path only better by)	0.398	
Welch t	-1.97	

the encoding redundancy does not vanish at longer sequences, while acknowledging the weaker statistical strength at this scale.

D. Comparison with Pixel-Space PixelCNN++

We re-implement PixelCNN++ [2] with the same CLIP-FiLM conditioning, the same DMoL head, and matched parameter count (6 417 795, 98.4% of our model). Critically, the evaluation function is identical: per-image NLL summed over all pixel positions, divided by $HWC \ln 2$.

Table III reports the comparison. PixelCNN++ achieves a substantially better absolute likelihood on this dataset: the gap to our best signal-space configuration is roughly 1.7 bpd at 30 epochs and 0.7 bpd at 60 epochs. Figure 3 shows generated images from both models on the same six text prompts.

Two qualifications are in order. First, the signal-space model’s effective sequence is *longer* than the pixel-space model’s by a factor of $2 \cdot (1 + f) \approx 2.16$, so per-step the comparison is less unfavourable than it appears. Second, the path-encoding effect we report is internal to the signal-space architecture and does not generalise to pixel-space: PixelCNN++ has no separate “path” to encode.

E. Negative Results: CRT-Physics Modifications

To check that the path-encoding result is not an isolated artefact within a broader “CRT-inspired” framing, we test three additional modifications motivated by the analog physics of cathode-ray tubes: **P1, learned beam focus**: an auxiliary head emits a per-step $\sigma_i \in [0.3, 1.5]$ which the renderer uses in place of the fixed σ . **P2, phosphor persistence**: a learnable

TABLE III
DIRECT COMPARISON ON OXFORD FLOWERS 32×32 AT MATCHED PARAMETER COUNT. MULTI-SEED NUMBERS ARE MEAN $\pm$ STD ACROSS SEEDS 0–2; SINGLE-SEED ENTRIES USE SEED 0.

Model	Params (M)	Epochs	bpd (test)
<i>Signal-space (this work)</i>			
full	6.52	30	7.156 ± 0.056
path only	6.52	30	6.432 ± 0.183
path only	6.52	60	5.24 ^a
<i>Pixel-space baseline</i>			
PixelCNN++	6.42	30	4.709 ± 0.022
PixelCNN++	6.42	60	4.58 ^a

^a Single seed.

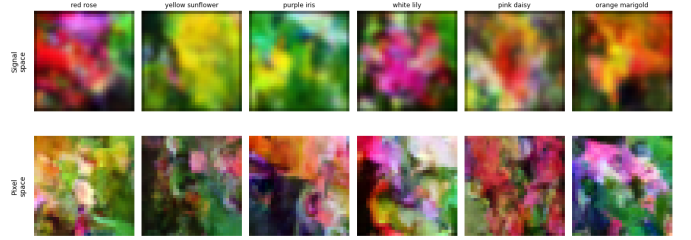


Fig. 3. Random samples (temperature 1.0) from the best-checkpoint signal-space model (path only, 60 epochs, top row) and PixelCNN++ (60 epochs, bottom row) on six held-out text prompts. Both models were conditioned on the same CLIP ViT-B/32 embeddings. Oxford Flowers 32×32 .

convex combination of the rendered image with a 3×3 blur of itself, initialised at 10% blur weight. **P3, beam-position perturbation**: an auxiliary head emits per-step $(\Delta x_i, \Delta y_i) \in [-0.5, 0.5]$ pixels added to the canonical path position, with a small ℓ_2 penalty on Δ .

Each modification is tested independently against a path-only baseline at three seeds for ten epochs. Table IV reports the result under a Welch t -test.

All three modifications produce $|t| < 1$. We report this as a meaningful negative result: the tested CRT-physics framing does not generalise to additional architectural choices. The path-encoding finding stands alone within this paper’s design space.

V. EXTENSIONS: TRAVERSAL LOCALITY AND ARCHITECTURAL PROBES

Two follow-up questions arise from the path-encoding result of Section IV-B. First, if 2D path geometry matters, does the *choice* of 2D path matter among well-designed alternatives? Second, can architectural modifications close the gap to pixel-space methods? We answer each in turn – both negatively in absolute terms, but informatively about the underlying bottleneck.

For this analysis we use $N=1024$ (single-sample-per-pixel raster, no flyback) to enable a fair comparison across traversals with the same sequence length. This differs from the V5 setup of Section IV-B ($N=2208$, 2-spp + flyback); the two are not directly comparable, and we report only same- N deltas here.

TABLE IV

CRT-PHYSICS MODIFICATIONS VS. THE PATH-ONLY BASELINE ON OXFORD FLOWERS 32×32 , 10 EPOCHS, 3 SEEDS. NONE REACHES SIGNIFICANCE.

	NLL mean	NLL std	Δ vs. base	t
baseline (path only)	17 862.9	829.8		
P1, learned σ	17 472.4	185.6	-390.5	-0.80
P2, persistence	17 953.8	910.5	90.9	0.13
P3, beam perturb	17 499.1	266.7	-363.8	-0.72

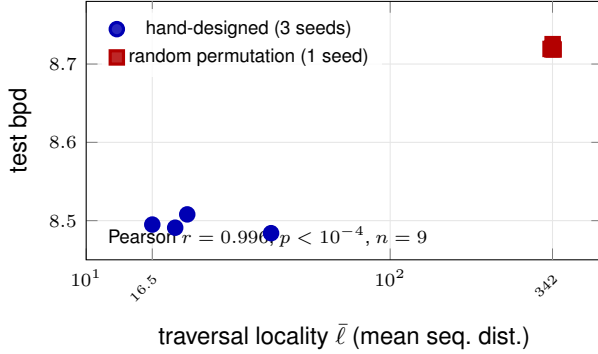


Fig. 4. Traversal locality $\bar{\ell}$ (mean sequence distance between four-neighbour pixel pairs) vs. test bpd on Oxford Flowers 32×32 , $N=1024$, 30 epochs. Hand-designed paths (blue circles) cluster at $\bar{\ell} \in [16.5, 40.6]$ and bpd $\in [8.48, 8.51]$, which is statistically indistinguishable. Random pixel permutations (red squares) cluster at $\bar{\ell} \approx 342$ and bpd ≈ 8.72 . Across the full $20\times$ range the relationship is nearly linear in $\log \bar{\ell}$ (Pearson $r=0.996$, $p < 10^{-4}$, $n=9$), identifying 1D-serialisation of 2D structure as the dominant bottleneck.

A. Locality Predicts bpd Across Traversals

We define a traversal’s *locality* $\bar{\ell}(\mathcal{P})$ as the mean sequence distance between the four-neighbour pairs of spatially adjacent pixels under path $\mathcal{P}$. We compare four hand-designed traversals (raster, Hilbert, diagonal, spiral) and five random pixel permutations, training each with the same path-only configuration as Section IV-B.

Table V reports test bpd alongside $\bar{\ell}$. Within the hand-designed paths, bpd is statistically indistinguishable: all four lie within ± 0.013 bpd, well inside seed noise. Across the full range of $\bar{\ell}$ – from 16.5 (raster) to ≈ 342 (random permutations) – bpd grows almost linearly with $\bar{\ell}$, with Pearson $r=0.996$ ($p < 10^{-4}$, $n=9$; Fig. 4). Locality is therefore *not* predictive among well-designed paths, but is a remarkably tight predictor across the full range, locating the bottleneck in 1D serialisation of 2D-neighbour structure rather than the specific choice of scan order.

B. Architectural Probes Do Not Close the Gap

Motivated by the locality finding, we test four interventions designed either to improve locality (patch-hierarchical traversal), to inject local inductive bias along the sequence (causal 1D conv-stem), to diversify the scan-order seen during training (epoch-level traversal augmentation), or to re-parameterise the predictive target (delta input encoding). Table VI reports bpd against the matched- N raster baseline (8.495 bpd, 3 seeds).

TABLE V

TRAVERSAL LOCALITY VS. TEST BPD AT $N=1024$, 30 EPOCHS. HAND-DESIGNED PATHS USE 3 SEEDS; RANDOM PATHS USE 1 SEED EACH. LOWER BPD IS BETTER.

Traversal	$\bar{\ell}$	bpd (test)
raster ($N=1024$)	16.5	8.495 ± 0.023
Hilbert	19.6	8.491 ± 0.023
diagonal	21.5	8.508 ± 0.022
spiral	40.6	8.484 ± 0.024
random r_0	342.1	8.720
random r_1	341.3	8.719
random r_2	345.0	8.719
random r_3	342.3	8.725
random r_4	337.0	8.719
Pearson $r = 0.996$, $p < 10^{-4}$, $n=9$		

TABLE VI

ARCHITECTURAL PROBES AGAINST THE RASTER BASELINE AT $N=1024$, 30 EPOCHS. NONE IMPROVES OVER BASELINE; THE CONV-STEM SEED-0 RUN TRIGGERED AN EARLY-ABORT GATE (SEED-0 BPD $>$ BASELINE $+ 0.30$).

Variant	bpd mean	Δ baseline	seeds
raster baseline	8.495 ± 0.023	—	3
delta input encoding	8.768 ± 0.014	0.273	3
patch-hierarchical (8×8)	8.512 ± 0.022	0.017	3
traversal augmentation	8.505 ± 0.026	0.010	3
causal 1D conv-stem	8.799	0.304	1

None of the four probes improves over the baseline. Three (patch-hierarchical, traversal augmentation, delta encoding) lie within or just outside seed noise and may be benign reorderings of the same problem; the conv-stem is worse by 0.30 bpd, sufficient to trigger early abort. We read these results as narrowing the design space: short-range temporal aggregation by a causal 1D conv-stem is the wrong inductive bias (the bottleneck demands 2D-neighbour aggregation), and surface-level reorderings of the path do not alter the underlying 1D-serialisation problem identified in Section V-A. The remaining gap to pixel-space methods most likely requires either a 2D-aware computational block (e.g. local 2D-window attention with appropriate causal masking, or a patch-level transformer with intra-patch and inter-patch attention) or a fundamentally different output factorisation. We leave both directions to future work.

VI. DISCUSSION

a) *Why does the sequence encoding hurt?*: Both sinusoidal encodings carry the same Shannon information about position: the path is invertible, so the sequence index and the 2D coordinate determine each other. The two encodings differ in *geometry*: the sequence encoding lays positions on a single 1D line, while the path encoding lays them on a folded 2D trajectory whose neighbours in path space are also neighbours in sequence space, but whose far-apart points in path space can be near-neighbours in 2D after the row wraps. We conjecture (without proving) that adding both encodings creates competing gradient signals on a shared embedding subspace and slows the

optimiser without providing additional information. A more careful study of the learned embeddings in each configuration is left to future work.

b) Why doesn't this close the gap to PixelCNN++?:

Pixel-space autoregressive models exploit local 2D conditional structure directly through their masked convolutions. The signal-space model has to recover this structure through a 1D causal attention over a sequence that visits 2D-neighbours across long sequence distances (one full row apart). Even with the correct positional encoding, this is a harder optimisation problem for the same parameter budget. We tested four candidate interventions in Section V-B – including a patch-hierarchical scan inspired by space-filling curves [13], [12] – and none improved over the matched- N baseline. Combined with the locality finding of Section V-A, this points specifically to 2D-aware intermediate computations (rather than further scan-order exploration) as the remaining lever; we leave that to future work.

c) Limitations.: (i) All experiments are on a single dataset (Oxford Flowers 102) at small resolutions. Generalisation to natural-image distributions such as CIFAR-10 or LSUN is not tested. (ii) The 48×48 scaling result shows the correct direction but with Welch $t = -1.97$, weaker than the 32×32 result ($t = 5.4$), due to high cross-seed variance; one seed achieved unusually low loss while the others converged to a different basin. (iii) The transformer's $O(N^2)$ attention prevents experiments at $\geq 64 \times 64$ in our compute budget without architectural changes such as KV caching or sparse attention. (iv) The PixelCNN++ implementation uses a single causal stack rather than the original's dual horizontal-and-vertical stacks; published PixelCNN++ numbers on CIFAR-10 (2.92 bpd) are therefore not directly attainable in our setup.

VII. CONCLUSION

In raster-scan signal-space autoregressive image generation, an explicit 2D path positional encoding makes the standard 1D sequence encoding redundant, and adding both reliably degrades test likelihood. Three CRT-physics-motivated modifications failed to produce significant gains, suggesting that the encoding effect is specific rather than a symptom of broader architectural under-fit. A matched-parameter PixelCNN++ retains an absolute likelihood advantage on the same data, framing our finding as a clean ablation result within signal-space autoregressive models rather than a challenge to pixel-space methods. Extending these results, traversal locality predicts test bpd with Pearson $r = 0.996$ ($p < 10^{-4}$, $n=9$) across hand-designed and random paths at matched $N=1024$, and four architectural probes (delta input encoding, patch-hierarchical traversal, traversal augmentation, causal 1D conv-stem) all fail to improve over that baseline, locating the remaining gap to pixel-space methods in 2D-aware computation rather than scan-order choice.

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